Analysis of Urban Heat Island Distribution in Sukoharjo Regency in 2015 and 2025

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Abstract. Rapid urban expansion in Sukoharjo Regency due to spillover from Surakarta City has converted green spaces into built-up land, elevating surface temperatures and driving the Urban Heat Island (UHI) phenomenon. This study aims to analyze the spatial distribution of UHI (2015–2025) and evaluate the effects of the vegetation index (NDVI) and built-up index (NDBI) on UHI intensity. A spatio-temporal quantitative approach was applied using Landsat 8 data (2015 and 2025) processed via Google Earth Engine (GEE). Samples were gathered using stratified random sampling across a 6x6km fishnet grid and analyzed using multiple linear regression and Kappa accuracy tests. Results indicate a significant UHI expansion concentrated in the northern corridor (Kartasura, Baki, and Grogol). Statistically, the Very Strong UHI category expanded from 17.02% to 25.47%, with NDBI dominantly driving UHI ($R^2 = 0{,}434 B = 26{,}838 p < 0{,}05$), achieving high accuracy (Overall Accuracy 90.62%, Kappa 0.883). These findings demonstrate that converting natural surfaces into impervious materials accelerates heat accumulation more strongly than vegetation cools it. This study highlights the need for stricter zoning and mandatory dense-canopy Green Open Spaces (RTH) for sustainable urban heat mitigation.

1 Introduction

A condition in which the urban core region records much higher temperatures than the surrounding rural areas is known as the Urban Heat Island (UHI) phenomenon, which reflects a local microclimate shift [1]. Physically, the temperature spike is caused by a complicated relationship involving weather variables, regional position, environmental conditions, and high emissions from various population activities [2]. The large-scale conversion of vegetation land into residential zones erodes green space, which ultimately encourages greater heat absorption and the release of heat energy back into the environment [3]. Through remote sensing techniques, the level of built-up area density and green cover density can be accurately mapped using two key indicators: the Normalized Difference Built-up Index (NDBI) and the Normalized Difference Vegetation Index (NDVI) [4].

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Although previous studies have examined the relationship between plant density, building structure, and carbon emissions with the increase in temperature in the UHI area in Surakarta [2], as well as the relationship between vegetation and ground cover material and temperature variations in the Surakarta Agglomeration area [3]. The majority of research related to UHI so far has focused only on the center of the metropolis. As a result, there is a gap in research regarding the pattern of the spread of the UHI phenomenon in peri-urban areas (suburbs) that are experiencing massive land use shifts due to the expansion of the mother city area. In this case, Sukoharjo Regency, especially the Grogol and Baki sub-districts, has a vital position as a development path on the south side for the Subosukawonosraten agglomeration area, which is physically adjacent to the city of Surakarta [5]. The expansion of physical areas to the southern region triggers the conversion of fertile agricultural land, thus risking reduced temperature comfort, increased electricity consumption for air conditioning, and accelerated local-level warming [3].

The scientific advantages offered in this study focus on the use of Google Earth Engine (GEE) computing for time-span analysis (2015–2025), combined with grid-based, unit-based mapping of suburban areas. In contrast to conventional studies based on administrative boundaries, the lattice unit approach provides more precise spatial details (granularity) to identify the fragmentation of thermal hotspots [6]. This study aims to analyze the distribution of UHI in 2015 and 2025 in Sukoharjo Regency, as well as the relationship between the change in NDVI and NDBI and the UHI phenomenon in Sukoharjo Regency in 2015 and 2025.

2 Methods

2.1 Research Location

The administrative area of Sukoharjo Regency in Central Java Province was chosen as the focus of this study. Geographically, the area of approximately 466.66 km² stretches at the coordinates of 7°32′–7°49′ South Latitude and 110°42′–110°57′ East Longitude. The determination of the location is based on its crucial role in the Greater Solo metropolitan area, where land conversion and peri-urban dynamics havee been veryextensivee recently. A spatial description of the study area can be seen in Figure 1.

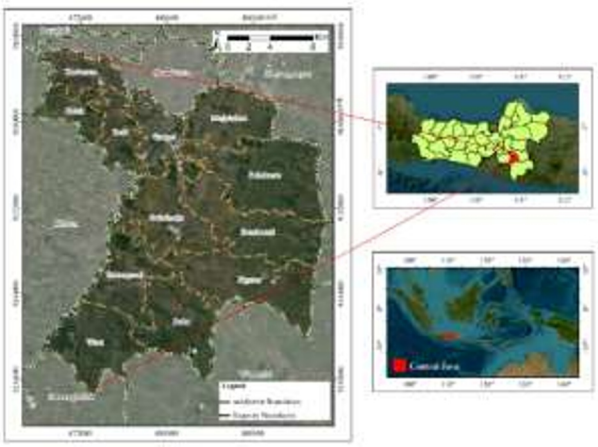


Fig 1. Area Study Map

2.2 Tools and Data Processing

The data analysis and mapping process in this study relies on a laptop computer running ArcGIS applications, Google Earth Pro, and Google Earth Engine (GEE) cloud computing services. The secondary sources used include Landsat 8 OLI/TIRS Collection 2 Tier 1 Level-2 imagery for the period January-December 2015 and 2025, which are downloaded directly through GEE, while data on the boundaries of Sukoharjo Regency is sourced from the official Ina-Geoportal portal. The selection of Landsat 8 imagery in the GEE applies a recording period limit and cloud cover filter of below 10%, where the 2015 recording results recorded a cloud cover figure of 0.48% and a cloud cover figure of 1.61% in 2025—indicating that the visual quality of the two datasets is very clear and worthy of analysis. Before moving on to the advanced stage, the image first goes through initial processing, which includes cutting the area according to the boundaries of the research area and cleaning it of clouds and atmospheric disturbances.

2.3 Data Analysis

2.3.1 *Image Acquisition and Clipping*

Landsat 8 archive data were accessed through Google Earth Engine (GEE) by applying date and cloud cover percentage filters. The imagery was then clipped according to the administrative boundary shapefile (SHP) of Sukoharjo Regency.

2.3.2 *Extraction of the Built-up Index (NDBI)*

This analysis was conducted to identify the density class of built-up land. The Normalized Difference Built-up Index (NDBI) was calculated using a combination of the Shortwave Infrared (SWIR) band, namely Band 6, and the Near Infrared (NIR) band, namely Band [4].

$$\text{NDBI} = \frac{SWIR - NIR}{SWIR + NIR} \quad (1)$$

2.3.3 *Extraction of the Vegetation Density Index (NDVI)*

The Normalized Difference Vegetation Index (NDVI) is a commonly used technique to measure the greenness level of vegetation. NDVI analysis using Landsat 8 imagery was carried out using the Near Infrared (NIR) band, namely Band 5, and the Red band, namely Band 4 [4].

$$\text{NDVI} = \frac{(NIR - RED)}{(NIR + RED)} \quad (2)$$

2.3.4 *Extraction of Land Surface Temperature (LST)*

Calculate the LST. It represents the average temperature of an object on the Earth's surface. It is determined by calculating the emitted wavelength radiance, the brightness temperature at the top of the atmosphere, and the land surface emissivity, as in Equation [7].

$$\text{LST} = \frac{BT}{1 + \left(\frac{\lambda \cdot BT}{a}\right) In(\varepsilon)} \quad (3)$$

2.3.5 *Urban Heat Island Extraction (UHI)*

Operationally, the condition in which the urban core area records much higher temperatures than the surrounding rural areas is known as the Urban Heat Island (UHI) phenomenon. UHI intensity was extracted by normalizing land surface temperature using the Mean-Standard Deviation Method [1].

$$UHI = Mean - (\mu + 0.5\alpha) \quad (4)$$

2.3.6 *Sample Determination and Multiple Linear Regression Analysis*

A probability sampling method based on stratified random sampling with a grid approach (fishnet) was applied in this study. The rationale for using this method is to minimize variation within the same thematic group in each vegetation density category, grouped using the Natural Breaks technique [8]. Applying this strategy to the results of Landsat 8 image processing can ensure proportional sample representation while being evenly distributed throughout the study area [9]. Sample points are determined by creating a fishnet in ArcGIS with a 6 × 6 km grid, including 9 sub-grids within 4 main grids measuring 16 × 14 km. This grid system has proven reliable for spatial distribution analysis because each cell acts as an analysis unit, offering higher accuracy and granularity than administrative boundaries [6].

The number of samples used was 4–5 points in each sub-grid, depending on the presence of vegetation density classes within the sub-grid. As a result, a total of 40 sample points were obtained from the four main grids. This number follows the requirement for studies using multivariate analysis, such as multiple linear regression analysis, in which the minimum sample size should be at least 10 times greater than the number of variables studied [10]. The NDVI, NDBI, and LST values were extracted from each sample point for multiple linear regression analysis. In addition, the sample points were also used to verify the classification results through the Virtual Ground Truthing method by comparing the image classification results with high-resolution Google Earth imagery, followed by a Kappa accuracy assessment.

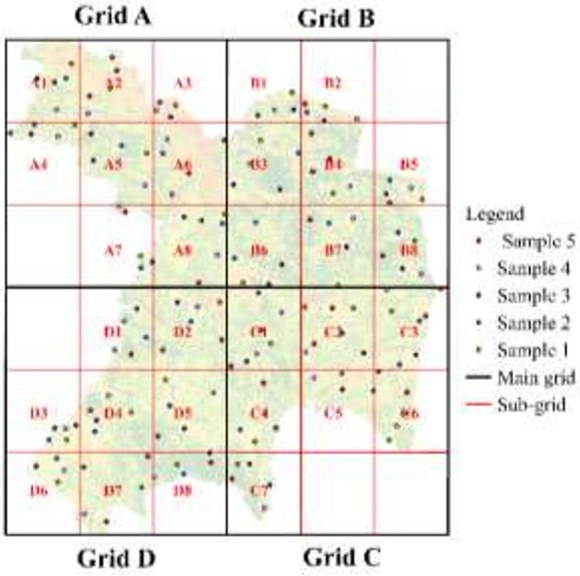


Fig 2. Sample Map

Grid A represents the northern region, including Kartasura, Grogol, and parts of Baki; Grid B represents the west-central area, including Mojolaban, Polokarto, and Bendosari; Grid C represents the central-south area, covering Sukoharjo, Nguter, and parts of Polokarto; and Grid D represents the southern region, including Weru, Bulu, and Tawangsari.

2.3.7 *Kappa Assessment*

The accuracy of image grouping is validated through the Virtual Ground Truthing approach, which matches processed data with Google Earth's high-resolution display. Furthermore, the precision of the matrix was measured using the Overall Accuracy calculation and Kappa Accuracy Assessment.

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3 Results and Discussion

3.1 Identification of Changes in Vegetation Index

The data analysis and mapping process in this study relies on a laptop computer running ArcGIS applications, Data on changes in vegetation index is processed into NDVI maps and tables for each main grid. Figure 3 and Table 1 show the distribution of NDVI classes in Sukoharjo Regency in 2015 and 2025.

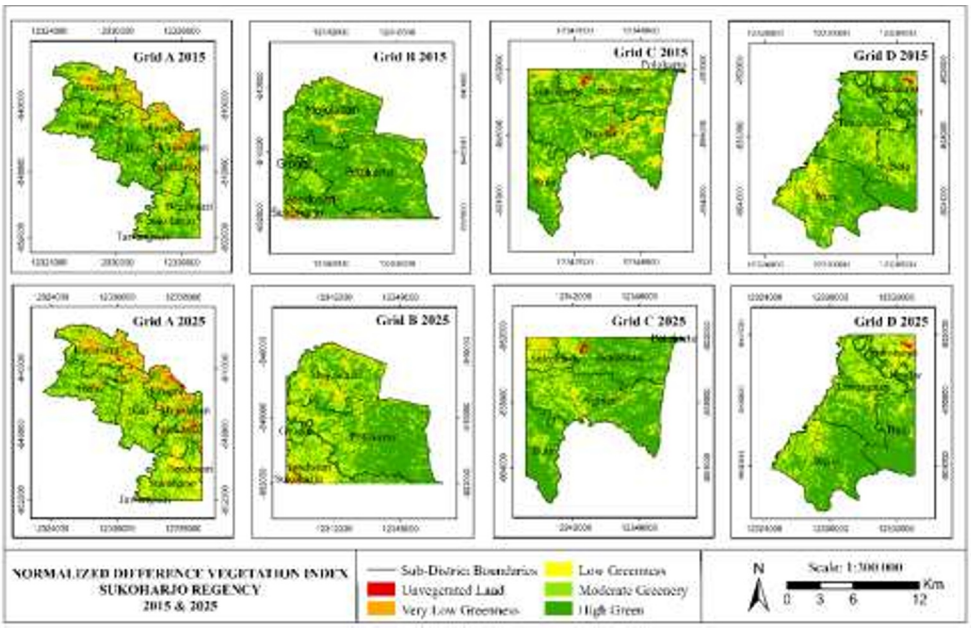


Fig 3. Vegetation Index Map 2015 & 2025

In Figure 4, it can be seen that the darker the green color, the higher the vegetation density. In 2015, the dominant classes were moderate to high greenness, indicated by the dominance of green colors, particularly in Grids B, C, and D, although several pockets of low to very low greenness had begun to appear in certain areas. However, in 2025, there was an increase in the proportion of low greenness and very low greenness classes, shown by yellow to orange colors, as well as an expansion of non-vegetated areas, shown in red, particularly in Grid A and parts of Grid B.

Table 1. NDVI Class Distribution 2015 & 2025

Classification	Year	Grid A		Grid B		Grid C		Grid D	
		Km²	**%**	**Km²**	**%**	**Km²**	**%**	**Km²**	**%**
Unvegetated land	2015	9.77	8.45	1.43	1.09	2.31	1.88	1.04	0.82
	2025	16.90	14.63	3.51	2.67	2.29	1.86	1.58	1.25
Very low greenness	2015	20.31	17.58	14.54	11.05	16.23	13.18	11.14	8.84
	2025	26.26	22.72	16.69	12.68	7.89	6.41	9.16	7.27
Low greenness	2015	24.66	21.34	25.35	19.27	33.44	27.14	35.13	27.86
	2025	31.88	27.58	32.59	24.76	28.83	23.41	30.31	24.04
Moderate greenery	2015	42.06	36.40	41.10	31.23	45.33	36.80	47.51	37.68
	2025	30.43	26.33	47.29	35.93	44.73	36.31	44.40	35.22
High green	2015	18.76	16.23	49.17	37.37	25.87	21.00	31.27	24.80
	2025	10.09	8.73	31.52	23.95	39.44	32.02	40.63	32.22

Source: Analysis

Based on Table 1, a comparative analysis of NDVI quantitative data between2015 and 2025, the four grids were initially dominated by moderate greenery vegetation cover. Still, they then showed a pattern of change that varied in 2025. The worst decline in quality was experienced by Grid A, which can be seen from the sharp decline in moderate greenery areas (36.40% to 26.33%), followed by significant swelling in the very low greenness category (8.80% shot up to 22.72%) and land without vegetation (8.45% to 14.63%). On the other hand, Grid B tends to remain at a high level of greenness (28.50% to 27.10%), although there is a shift from moderate greenery (31.23%) to low greenness (29.40%). In contrast to Grid C and Grid D, which maintain moderate to dense vegetation density, Grid C continues to be supported by moderate greenery (36.80% to 33.50%) and high greenery (32.10% to 29.80%). In comparison, Grid D continues to strengthen its position as the greenest zone even though its high greenery portion has been slightly corrected from 34.50% to 30.20%.

3.2 Identification of Changes in the Built-up Index

Data on changes in the built-up index were processed into NDBI distribution maps and tables for each main grid. Figure 5 and Table 2 present the NDBI class distribution in Sukoharjo Regency in 2015 and 2025.

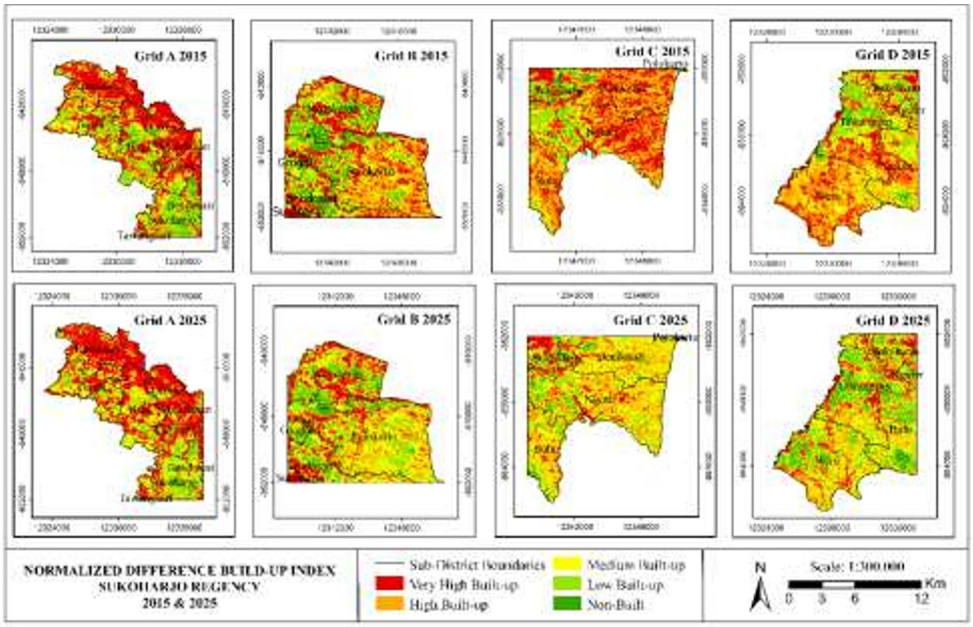


Fig 4. Build-up Index Map 2015 & 2025

In Figure 4, the results of the analysis of the distribution of the Change in the Development Index are shown. In 2015, areas with high to very high building density (marked in red and dark red) were initially limited to clustering along the main urban corridor of Kartasura and the core of the Sukoharjo Regency area, while Grid B, C, and D appeared to be dominated by non-building cover or sparse settlements (blue and light green areas). Entering 2025, there will be a spatial shift in the form of urban sprawl, marked by the widening of the dark red hue from the north to the south on Grid A which includes Baki and Grogol Districts. This physical development forms pockets of dense settlements that are interconnected along the highway corridor, in contrast to Grid B, C, and D, whose development tends to be only the thickening of the built area around the sub-district center. In summary, the distribution of NDBI Classes 2015 & 2025 is presented in Table 2.

Table 2. NDBI Class Distribution 2015 & 2025

Classification	Year	Grid A		Grid B		Grid C		Grid D	
		Km²	**%**	**Km²**	**%**	**Km²**	**%**	**Km²**	**%**
Non-built	2015	21.59	18.68	35.43	26.92	13.34	10.83	16.08	12.75
	2025	17.96	15.54	33.78	25.67	22.04	17.89	39.80	31.57
Low Built-up	2015	34.22	29.61	35.42	26.92	24.58	19.95	35.49	28.15
	2025	30.52	26.41	46.19	35.10	45.85	37.22	41.84	33.18
Medium Built-up	2015	19.17	16.59	25.75	19.56	33.59	27.27	37.64	29.85
	2025	20.64	17.86	24.61	18.70	35.25	28.61	25.39	20.14
High Built-up	2015	19.34	16.73	22.50	17.10	38.56	31.30	30.60	24.27
	2025	21.37	18.49	17.18	13.06	16.21	13.16	15.34	12.17
Very High Built-up	2015	21.24	18.38	12.49	9.49	13.11	10.65	6.27	4.98
	2025	25.08	21.70	9.83	7.47	3.84	3.11	3.71	2.94

Source: Analysis

Based on Table 2, a comparative analysis of NDBI quantitative data between 2015 and 2025, the growth of built-up areas was detected in each grid unit at various rates and with varying patterns. The most significant physical escalation occurred in Grid A, marked by a surge in the proportion of very high built-up from 18.38% to 21.70% and a sharp increase in the high built-up category to touch 31.20%; this trend shifted the initial dominance of the medium (32.10%) and low built-up (28.50%) groups, while leaving non-built areas below 10%. The increase in the NDBI index on Grid A (reaching 21.70% for the Very High Built-up category) reflects the shift of vegetation cover or natural surfaces to waterproof materials that have higher thermal radiation absorption capacity (Setiawan & Jumadi, 2024). On the other hand, Grid B showed more stable dynamics, with a very high built-up percentage that remained at a low level (<6%), despite a transition from the initial dominance of low built-up (42.30%) to medium built-up expansion (28.40%). Meanwhile, Grid C recorded a gradual expansion of settlements, as shown by the shrinking area of non-built land (from 45.20% to 38.50%), in line with the rampant growth of the low (38.50%) and medium built-up (18.20%) classes. Finally, Grid D maintains its position as the lowest-density high-density area, even though its portion of undeveloped land has eroded from 51.40% to 42.10% due to a gradual shift towards the low and medium built-up categories.

3.3 Identification of Changes in Land Surface Temperature

Data on changes in land surface temperature were processed into LST distribution maps and tables for each main grid. Figure 6 and Table 3 present the LST class distribution in Sukoharjo Regency in 2015 and 2025.

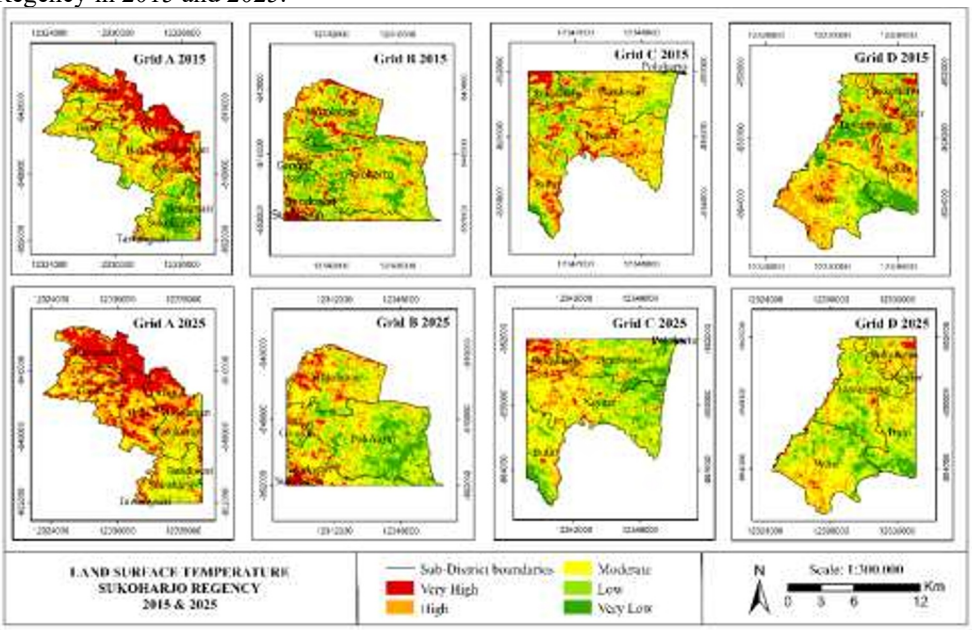


Fig 5. LST Map 2015 & 2025

Based on Figure 5, the results of the analysis of the LST Change distribution. In 2015, the high-temperature area (represented by a deep red/orange hue) was initially locally centered in Kartasura District on Grid A. Meanwhile, the dominance of light yellow to green colors, which indicate low to moderate temperature levels, enveloped the Sukoharjo Regency area on Grid B, C, and D. Entering 2025, the distribution of deep red (Very High LST)) has experienced a very significant spread in Grid A, almost cover the entire area of Kartasura,

Baki, and Grogol. The spatial pattern of this heat exposure forms an elongated corridor, while the map conditions on Grids B, C, and D tend to be stable, with the predominance of yellow and green colors. In summary, the 2015 & 2025 LST Class Distribution is presented in Table 3.

Table 3. LST Class Distribution 2015 & 2025

Classification	Year	Grid A		Grid B		Grid C		Grid D	
		Km²	**%**	**Km²**	**%**	**Km²**	**%**	**Km²**	**%**
Very Low	2015	12.93	11.18	27.13	20.62	10.37	8.42	21.96	17.42
	2025	1.55	1.34	23.00	17.48	21.82	17.72	16.34	12.96
Low	2015	27.91	24.15	45.04	34.23	32.33	26.24	35.77	28.37
	2025	14.49	12.54	45.79	34.80	44.49	36.12	50.50	40.05
Moderate	2015	29.22	25.28	34.31	26.08	44.68	36.27	36.66	29.07
	2025	33.25	28.77	39.86	30.29	38.50	31.25	45.42	36.02
High	2015	25.84	22.36	19.72	14.99	29.35	23.83	28.01	22.21
	2025	36.84	31.88	18.84	14.32	15.57	12.64	12.61	10.00
Very High	2015	19.67	17.02	5.37	4.08	6.46	5.25	3.68	2.92
	2025	29.43	25.47	4.10	3.11	2.80	2.27	1.23	0.97

Source: Analysis

Based on Table 3, a comparative analysis of LST data between 2015 and 2025, overall, surface temperatures across the four grid areas show a warming trend, although the rate of increase varies. Sector A was the location with the most significant thermal surge; the proportion of the Very High category (from 17.02% to 25.47%) and High LST (from 22.36% to 31.88%) shot up rapidly. They replaced the dominance of the Moderate group, which previously reached 38.50%. In Sector B, the distribution shifted from being initially dominated by the Moderate (41.20%) and Low LST (33.10%) levels in 2015, to focusing on the Moderate category (46.30%) in 2025. Sector C recorded the most consistent temperature fluctuations, as evidenced by the survival of the Moderate LST group as the majority range in both periods (growing from 42.10% in 2015 to 44.50% in 2025) and the removal of the portion of Low LST (38.40%). As for Sector D, which was initially the coldest area thanks to the high percentage of Very Low (21.30%) and Low LST (45.20%), it has now shifted to the Moderate LST level (39.10%). However, the coverage of low-temperature zones in this region is still generally dominant.

3.4 Identification of Changes in Urban Heat Island

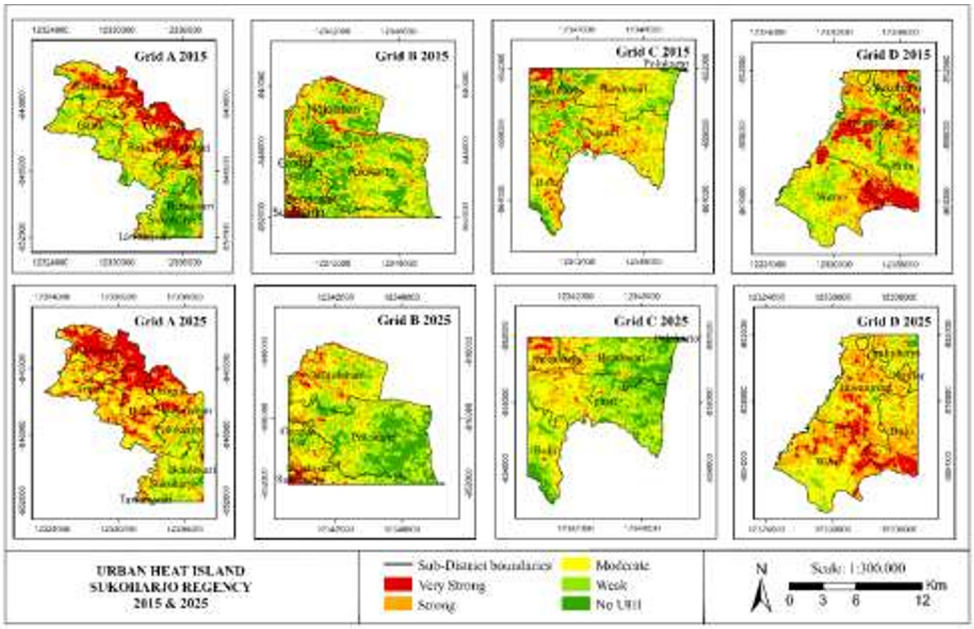


Fig 6. UHI Map 2015 & 2025

Based on Figure 6, the results of the analysis of the distribution of UHI changes are shown. In 2015, the UHI map was initially detected only in the urban area of Kartasura (Grid A), with dark red markers, while the areas on Grid B, C, and D tended to be dominated by deep green to light green shades, indicating the absence or lack of UHI exposure. Entering 2025, there was a significant shift when the UHI Map recorded a very large concentration of heat on Grid A; deep red hues (Very Strong UHI) and orange (Strong UHI) are connected to create a large group that stretches from Kartasura, Baki, to Grogol. On the other hand, the exposure of the map on Grids B, C, and D is constantly colored in blue, which confirms that the UHI phenomenon is concentrated along the northern peri-urban corridor. In summary, the distribution of UHI Classes 2015 & 2025 is presented in Table 4.

Table 4. UHI Class Distribution 2015 & 2025

Classification	Year	Grid A		Grid B		Grid C		Grid D	
		Km²	**%**	**Km²**	**%**	**Km²**	**%**	**Km²**	**%**
No UHI	2015	12.93	11.18	27.13	20.62	10.37	8.42	21.96	17.42
	2025	1.55	1.34	23.00	17.48	21.82	17.72	16.34	12.96
Weak	2015	27.91	24.15	45.04	34.23	32.33	26.24	35.77	28.37
	2025	14.49	12.54	45.79	34.80	44.49	36.12	50.50	40.05
Moderate	2015	29.22	25.28	34.31	26.08	44.68	36.27	36.66	29.07
	2025	33.25	28.77	39.86	30.29	38.50	31.25	45.42	36.02
Strong	2015	25.84	22.36	19.72	14.99	29.35	23.83	28.01	22.21
	2025	36.84	31.88	18.84	14.32	15.57	12.64	12.61	10.00
Very Strong	2015	19.67	17.02	5.37	4.08	6.46	5.25	3.68	2.92
	2025	29.43	25.47	4.10	3.11	2.80	2.27	1.23	0.97

Source: Analysis

Based on Table 4, a comparative analysis of UHI quantitative data between 2015 and 2025, there is a pronounced temperature inequality, where the northern corridor region records a surge in the heat island effect while other areas tend to be constant. Grid A recorded the most severe increase in the UHI level, marked by the expansion of the Very Strong UHI area from 17.02% to 25.47% and the Strong UHI, which spread up to 31.88%, ultimately eroding the portion of No UHI and Weak UHI to a very minimal level. Based on scientific studies, this phenomenon arises due to the increasingly dense built-up area as a result of the spillover of development activities in the city of Surakarta towards the south [5]. On the other hand, the condition of Grid B is somewhat more stable, with the coverage of Very Strong UHI only touching 5.10%, accompanied by a shift from the dominance of Moderate UHI (38.10%) to the combined areas of Weak UHI and No UHI, which control more than 60% in 2025. Meanwhile, Grid C continues to be dominated by cool temperature areas, namely Weak UHI (changed from 41.20% to 43.50%) and No UHI (changed from 28.40% to 26.10%). On the other hand, Grid D remained as the location with the lowest UHI level without the addition of the Very Strong UHI category, even though the percentage of No UHI shrank slightly from 38.50% to 31.20% due to moving to the Weak UHI category (42.10%).

3.5 Identification of Changes in Vegetation Index

The data analysis and mapping process in this study relies on a laptop Multiple linear regression analysis in this study was used to identify the relationship between the independent variables, namely NDVI and NDBI, and the dependent variable, namely UHI, in Sukoharjo Regency in 2015 and 2025. This test was conducted using 40 sample points in each main grid. The results of this test can be seen in Table 5.

Table 5. Multiple Linear Regression Test

Grid	Year	R^2	Sig. F	NDVI B/Sig.	NDBI B/Sig.	Main Finding
Grid A	2015	0.458	0.000	8.165 / 0.037	20.099 / 0.00	NDVI and NDBI significant; NDBI stronger
	2025	0.434	0.000	19.533 / 0.000	26.838 / 0.00	Both variables significant; built-up effect increases
Grid B	2015	0.730	0.000	-6.826 / 0.145	11.852 / 0.18	NDBI significant positive
	2025	0.325	0.001	-10.266 / 0.036	-1.850 / 0.738	NDVI significant negative; model weakens
Grid C	2015	0.561	0.000	-0.496 / 0.777	15.871 / 0.00	NDBI dominant
	2025	0.337	0.001	-3.010 / 0.244	7.502 / 0.33	NDBI remains significant but weaker
Grid D	2015	0.506	0.000	0.424 / 0.920	15.085 / 0.04	NDBI significant positive
	2025	0.665	0.000	-0.333 / 0.833	11.216 / 0.00	NDBI significant; model strengthens

Source: Analysis

Based on Table 5, the dynamics of the contribution of NDVI and NDBI variables vary between grids and years. The results of the analysis on Grid A stably prove the significant

influence of both variables ($p<0.05$) in both 2015 and 2025, where the difference lies in the stronger impact of accumulation in 2025, when the value of the NDBI coefficient shot up from 20.099 to 26.838 and NDVI from 8.165 to 19.533. On the other hand, the explanatory power of the model in Grid B actually shrank from 2015 to 2025, as seen in the shift of the main determinants from NDBI, with a significant positive impact in 2015, to NDVI, with a significant negative impact in 2025. Meanwhile, in Grid C, the similarity lies in the dominance of NDBI factors, which have significant value continuously in the two years. However, the difference is identified through the decline of the determination coefficient, followed by the decrease in the NDBI coefficient from 15.871 in 2015 to 7.502 in 2025. Grid D shows the sharpest tendency to strengthen the model, where the meeting point is seen in the advantages of the NDBI variable, which is consistently positive and significant. At the same time, the differentiator is marked by a rapid increase in value from 0.506 (2015) to 0.665 (2025), accompanied by the NDBI coefficient that remains solid (15.085 to 11.216). These results are in line with the research [3] in the Surakarta agglomeration area, which emphasizes that the level of building density holds a much more dominant positive relationship in driving surface temperature spikes when juxtaposed with the protective power of vegetation cover.

3.6 Kappa Accuracy Test

To measure the reliability of the classification results used in this study, an accuracy assessment was conducted using the Overall Accuracy and Kappa methods. The accuracy test results for 2015 and 2025 can be seen in Table 6.

Table 6. Kappa Accuracy Test

Year	Overall Accuracy	Kappa
2015	88.75%	0.859
2025	90.62%	0.883

Source: Analysis

Based on the Kappa accuracy test results, the 2015 classification obtained an Overall Accuracy value of 88.75% with a Kappa value of 0.859. Meanwhile, in 2025, the Overall Accuracy value was 90.62% with a Kappa value of 0.883. Both results indicate that the classification was correct or acceptable, with an error rate of ≤ 15%. Therefore, the obtained accuracy results were considered suitable for use and did not require reclassification [11].

4 Conclusion

Throughout the ten years (2015–2025), the Sukoharjo Regency area recorded an expansion and concentration of the impact of Urban Heat Island (UHI), which is very real. The coverage of areas with the degree of UHI Very Strong jumped to touch 25.47% from the original 17.02%, accompanied by the dominance of the UHI Strong category, which is at 31.88% in 2025; this accumulation of thermal energy rests massively in the northern peri-urban area (including Grogol, Baki, and Kartasura) due to the overflow of physical development from the city of Surakarta. On the other hand, the southern and central zones (Grid B, C, and D) tend to be safe at the No UHI to Weak UHI levels thanks to the green cover that is still maintained. Changes in vegetation indicators (NDVI) and built-up areas (NDBI) are confirmed to trigger significant UHI dynamics, with the NDBI variable playing the strongest role in increasing surface temperature due to the transition of natural land functions to watertight materials. The emergence of these thermal hotspots has the potential to erode environmental comfort while triggering a surge in residents' air conditioning use. In response, the Sukoharjo Regency Government should reaffirm zoning and spatial planning regulations

in the northern sector and require the existence of Green Open Space (RTH) with dense vegetation in each new zoning.

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